RTL FUNCTIONAL VERIFICATION / ICARUS VERILOG

6-TAP FIR FILTER

Simulation results • Seven test scenarios • Sample-by-sample comparison

SAMPLES CHECKED	PASSED	FAILED	TEST SCENARIOS
95	95	0	07

OBSERVED IMPULSE RESPONSE

h[n] = [2, 4, 6, 6, 4, 2]

Coefficient sum = 24

STEADY-STATE RESPONSE

20 → 480

255 → 6120

Constant input / Maximum input

SIGNAL KEY

– INPUT x[n]

··· OUTPUT y[n]

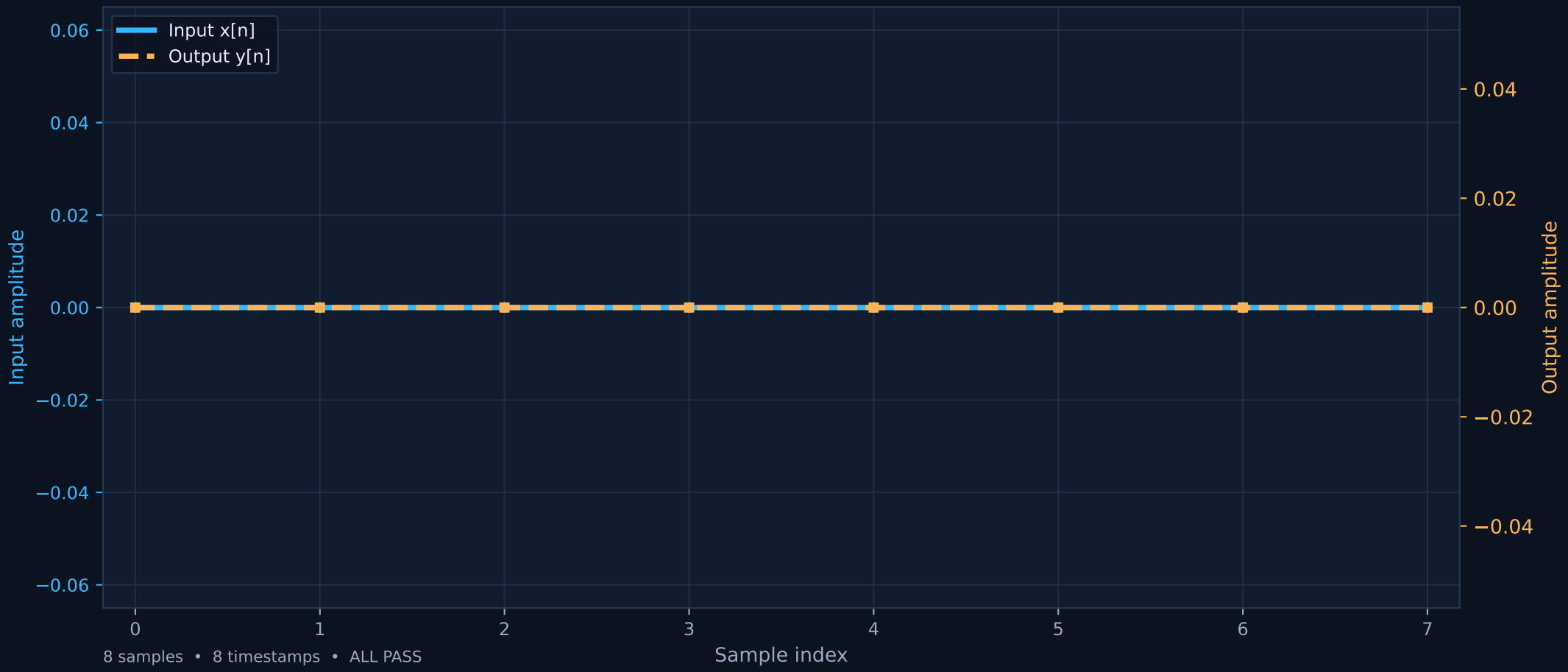
All 95 comparisons passed. No mismatches were reported by the supplied testbench.

Timestamps are reproduced from the simulation log; reported time unit is 1 ps.

TEST 01 / FUNCTIONAL RESPONSE

Zero input

Input and output shown together • Independent amplitude scales



8 samples • 8 timestamps • ALL PASS

OBSERVATION

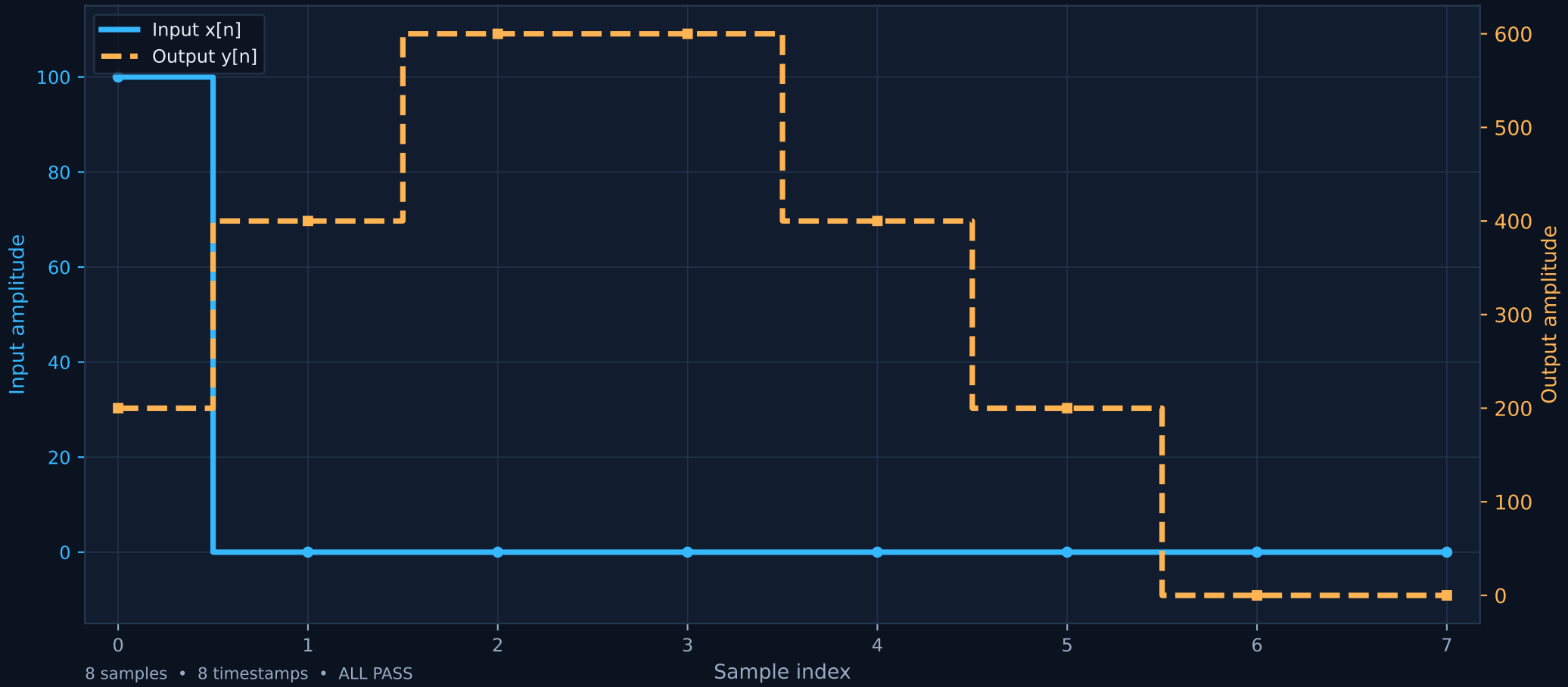
Zero input produces a zero output throughout the observation window.

Blue / left axis = input | Amber / right axis = filtered output

TEST 02 / FUNCTIONAL RESPONSE

Impulse response

Input and output shown together • Independent amplitude scales



OBSERVATION

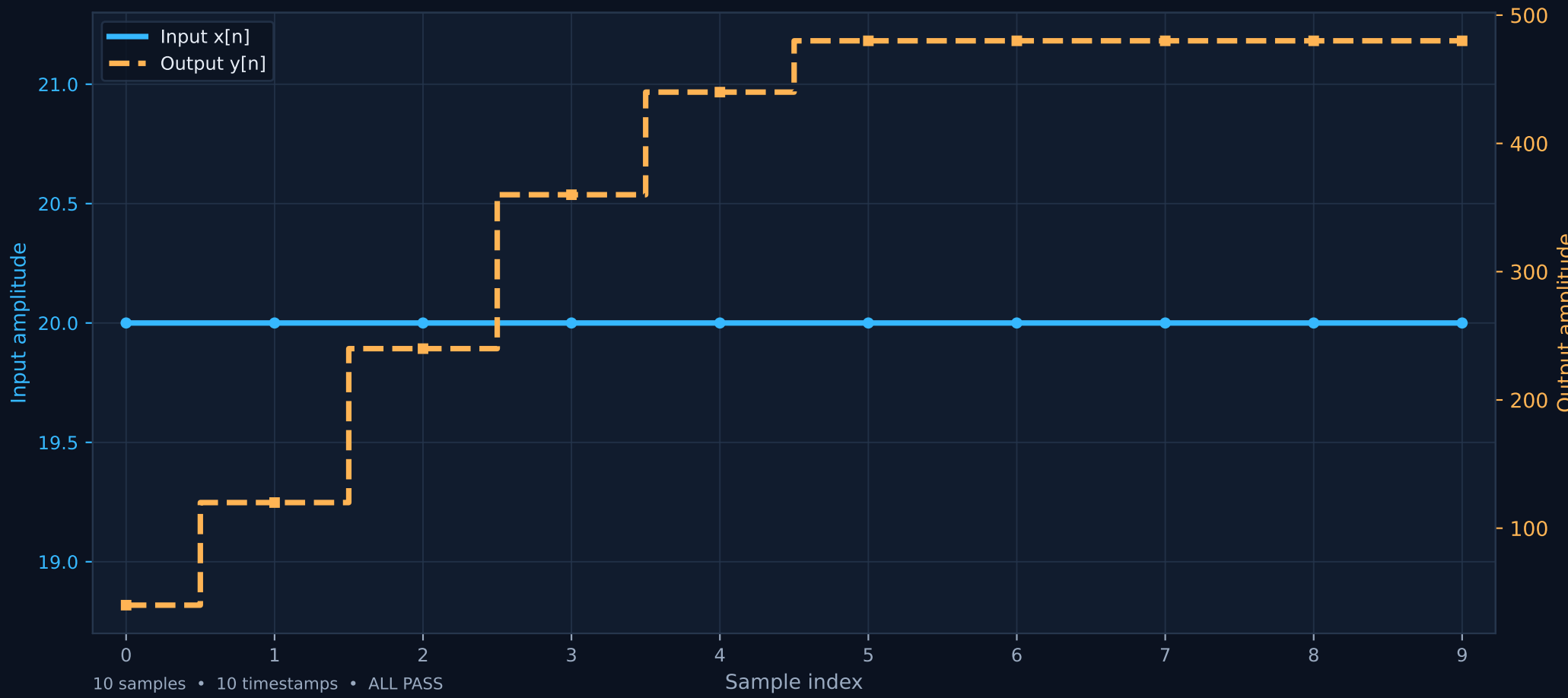
The impulse response exposes the six tap weights: 2, 4, 6, 6, 4, 2.

Blue / left axis = input | Amber / right axis = filtered output

TEST 03 / FUNCTIONAL RESPONSE

Constant input

Input and output shown together • Independent amplitude scales



OBSERVATION

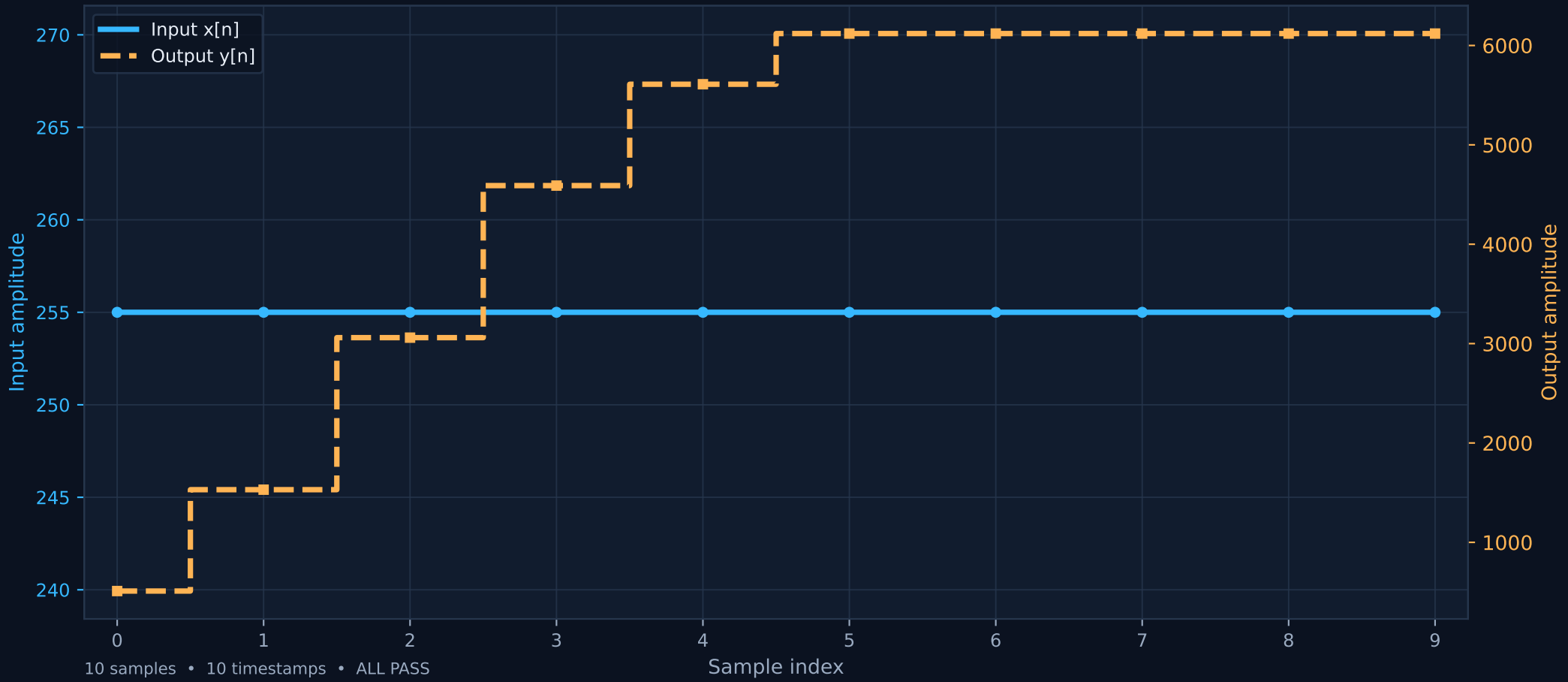
The output settles at 480, equal to input 20 multiplied by the coefficient sum 24.

Blue / left axis = input | Amber / right axis = filtered output

TEST 04 / FUNCTIONAL RESPONSE

Maximum input

Input and output shown together • Independent amplitude scales



OBSERVATION

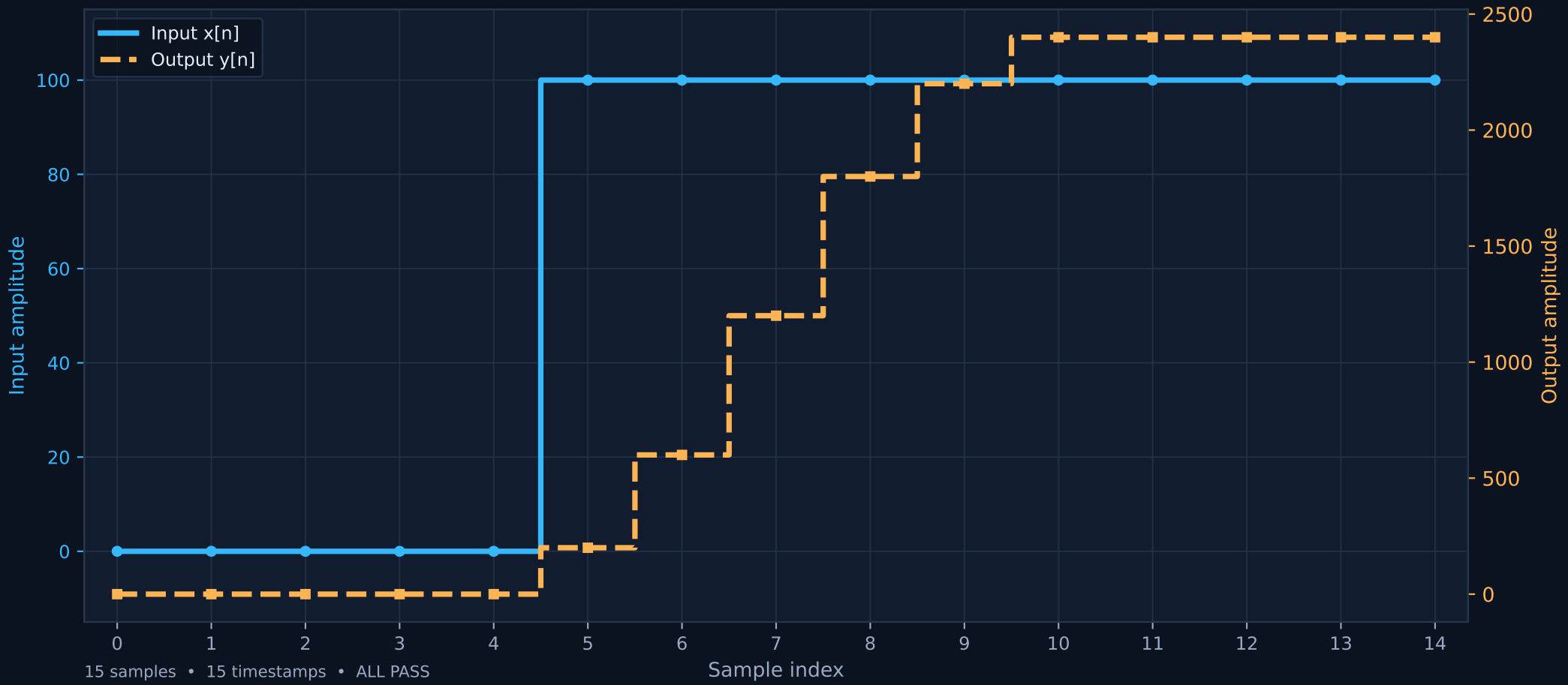
Full-scale input settles at 6120, equal to 255 multiplied by the coefficient sum 24.

Blue / left axis = input | Amber / right axis = filtered output

TEST 05 / FUNCTIONAL RESPONSE

Step input

Input and output shown together • Independent amplitude scales



OBSERVATION

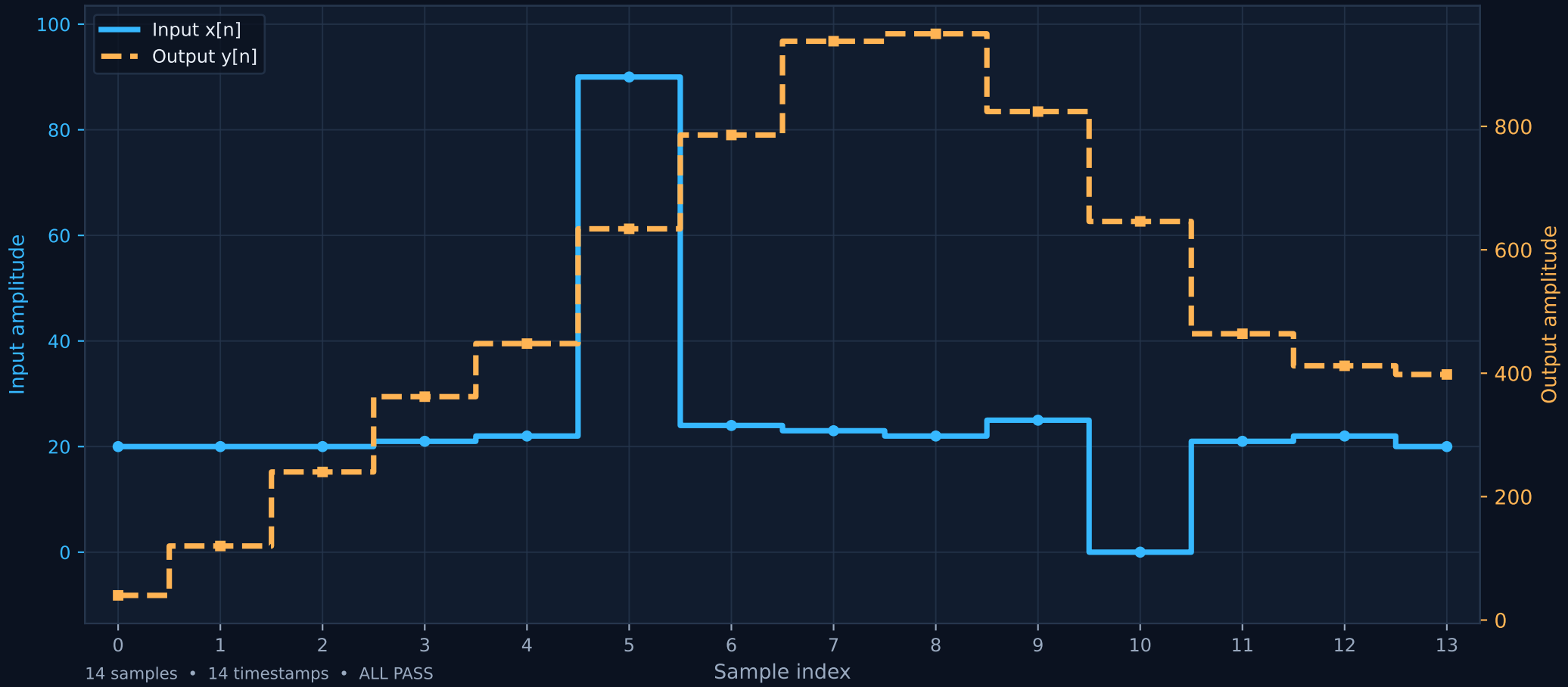
After the step from 0 to 100, the output builds over six samples and settles at 2400.

Blue / left axis = input | Amber / right axis = filtered output

TEST 06 / FUNCTIONAL RESPONSE

Varying input

Input and output shown together • Independent amplitude scales



14 samples • 14 timestamps • ALL PASS

OBSERVATION

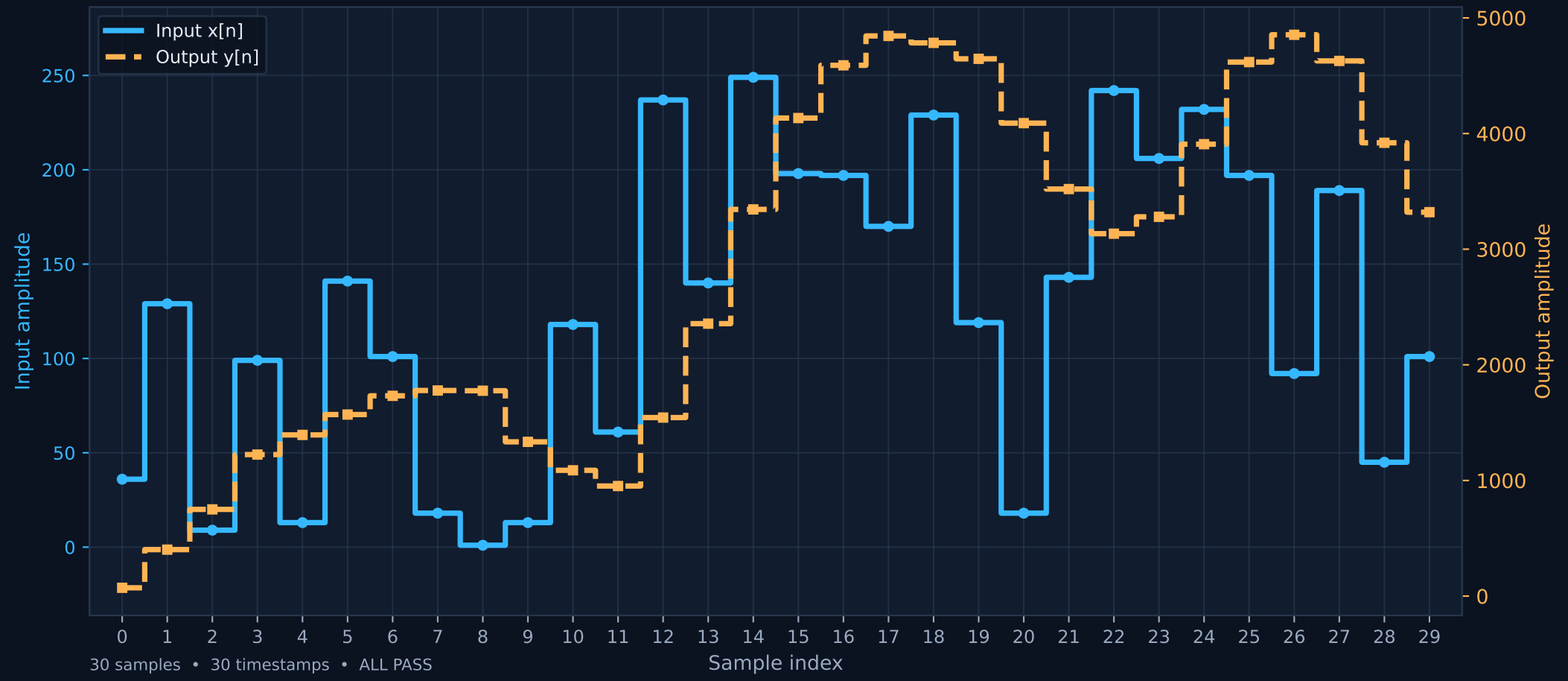
Each output depends on the current input and five preceding samples.

Blue / left axis = input | Amber / right axis = filtered output

TEST 07 / FUNCTIONAL RESPONSE

Random input

Input and output shown together • Independent amplitude scales



OBSERVATION

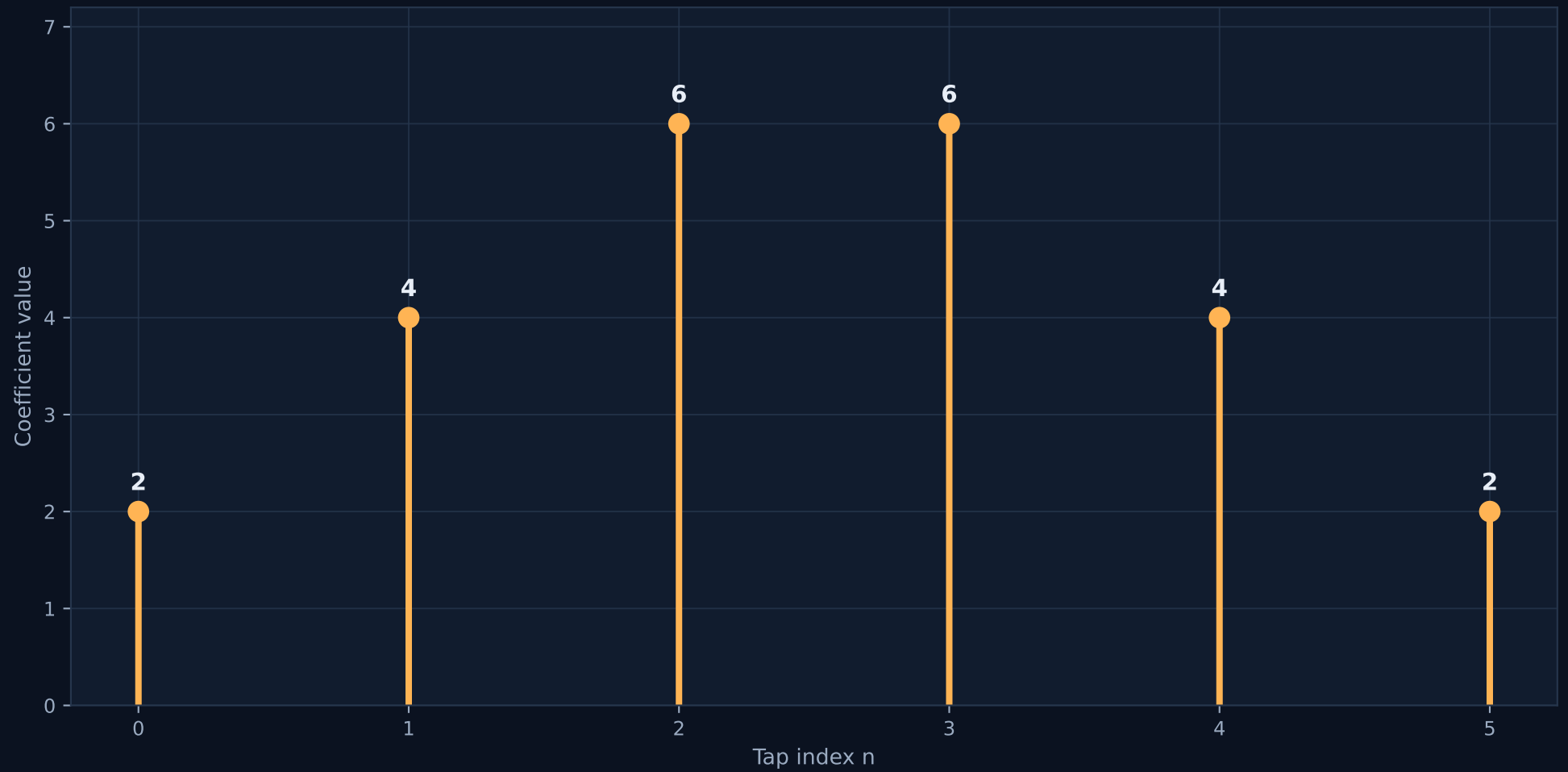
The output varies with the input history, showing the filter's response to a changing sequence.

Blue / left axis = input | Amber / right axis = filtered output

FILTER CHARACTERISATION

Impulse response & tap weights

A single non-zero input reveals the six-sample response of the FIR filter.



h[n] = [2, 4, 6, 6, 4, 2]

Sum of coefficients = 24 • Symmetric tap sequence